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Your details

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Assessment details

Course: DECO7230 Digital Prototyping and Extended Reality

Assignment: Design Concept Report

AI use

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ChatGPT	Used to check spelling and grammar and edit my final document	Ensure my report's grammatical accuracy	Entire work	06 October 2026

This coversheet has been adapted with thanks to BIOM3020 Integrated Endocrinology.

MiroXR: Embodied Sticky Note Organisation

Concept Overview

Prototype 2 is an XR-based interactive environment in which users interact with virtual sticky notes using a Meta Quest controller. Based on findings from the first design evaluation, the second prototype focuses on improving interaction clarity, reducing unnecessary complexity, and providing more direct spatial interaction.

The second evaluation investigates whether the refined XR interactions are understandable, usable, and reliable when experienced on an actual Meta Quest 3S device.

Evaluation Goals / Research Questions

RQ1 — Interaction
Can users understand how to select and interact with virtual notes using the Quest controller?

RQ2 — Usability
Can users complete the main interaction tasks with minimal errors or assistance?

RQ3 — Experience
Does the revised XR environment provide a clearer and more manageable experience than the previous prototype?

Research Method

Participants：4 participants were recruited for the evaluation (in week9 studio).

Equipment：Meta Quest 3S / Unity prototype / Quest controllers / Observation sheet / Post-test interviews

Procedure：1 Introduction + consent ；2 Brief explanation of the XR environment；3 Participant wears Quest 3S；4 Complete predefined tasks；5 Researcher observes performance；6 Short post-test interview/questionnaire

Think-aloud was not used as the primary method in Evaluation 2 because participants in the previous evaluation reported discomfort with continuously verbalising their actions. Instead, behavioural observation was combined with post-task interviews to reduce interference with the XR experience.

The Design Evaluation 2 usability testing was conducted during the Week 9 Studio session with four student participants.

Task

Task 1: Identify a note that can be interacted with.

Task 2: Select/pick up the target note using the controller.

Task 3: Move or interact with the note as intended.

Task 4: Release/complete the interaction without researcher assistance.

Success Criteria / Metrics

Metric	Target
Task completion rate	≥ 80%
Successful interaction without assistance	≥ 75%
Major errors	≤ 1 per participant
Researcher assistance	≤ 1 time per participant
Overall usability rating	≥ 3/4

Result

Four classmates participated in the evaluation during the Week 9 Studio. All participants completed all four tasks, giving an overall task completion rate of 100% (16/16), including tasks completed with assistance. Fifteen tasks were completed independently (93.75%), and three participants completed the entire task sequence without assistance. P02 initially pressed the grip button when attempting to select a note and required one prompt to use the index trigger. No other errors or researcher interventions were recorded. The mean usability rating was 4.75/5, with individual ratings of 5, 4, 5 and 5. All four participants completed the four tasks successfully, with only one participant requiring assistance during the selection task. Detailed records are provided in Appendix A, Figure A1.

Findings

Observation and interview records indicated that the interactive notes were easy to identify and that the existing visual feedback clearly communicated interaction states. The main observed usability issue concerned controller button mapping: P02 understood the interaction after a brief prompt but did not initially identify the correct button.

Discussion

These results suggest that the core interaction sequence was usable within this small studio sample. However, successful completion did not always indicate independent first-use understanding. P02's difficulty highlights an opportunity to make the required controller input more explicit. Placement and additional feedback suggestions represent opportunities for refinement despite successful task performance. The evaluation involved only four classmates, all with some previous VR experience, so the findings may not represent first-time VR users. These results alone also do not establish improvement over the previous prototype.

Proposed Design Changes

The next iteration will prioritise a contextual "Pull Trigger to Pick Up" hint to address the observed button confusion. Placement guidelines or optional soft snapping will be considered in response to alignment and depth-positioning feedback. Subtle haptic or audio cues will also be considered as supplementary confirmation. Follow-up testing will examine whether these changes improve independent selection and placement precision.

Appendix A Observation Data

Figure A1. Design Evaluation 2 observation results collected during the Week 9 Studio session.

This table summarises the observational data collected from four participants during the Week 9 Studio usability testing session using the Meta Quest 3S prototype. It records task completion, independent completion, errors, researcher assistance, observed behaviour, and overall performance across four interaction tasks. These observations were later analysed to identify usability issues and support the findings presented in the main report.

DECO7230 Design Evaluation 2 — Observation

Meta Quest 3S • Prototype 2 / 2A • Edit task wording to match the tested build before testing.
Enter VR experience once per participant. Pale yellow = input; pale blue = automatic summary. IDs are reusable slots, not collected data.
Completion: Yes = task goal achieved, even with help; No = attempted but not achieved. Leave untested tasks blank.
Without assistance: Yes only when completed with no help; otherwise No. Errors = observable incorrect actions; describe technical faults in Notes.
Assistance count = separate researcher prompts/interventions beyond the standard briefing. Enter 0 when none. Use the same rules for everyone.
After each participant, complete the Overall observation table below. Totals appear only after all four relevant task entries are recorded.

Participant ID	VR experience	Task	Completion	Without assistance	Errors	Assistance count	Observed behaviour	Notes
P01	Moderate	1. Identify an interactive note	Yes	Yes	0	0	Looked around environment, immediately identified floating note card within 3 seconds.	Clear visual affordance
P01		2. Select / pick up the target note	Yes	Yes	0	0	Aimed laser pointer directly at target note and pressed index trigger to select on first attempt.	Natural trigger selection
P01		3. Perform the intended note interaction	Yes	Yes	0	0	Smoothly moved and repositioned note in 3D space with continuous trigger hold.	Stable 6-DoF manipulation
P01		4. Release / place the note correctly	Yes	Yes	0	0	Released trigger cleanly to place note; observed anchored position.	Clean release and placement
P02	Limited	1. Identify an interactive note	Yes	Yes	0	0	Looked around for ~5 seconds and correctly identified the interactive note card.	Easily distinguished from background
P02		2. Select / pick up the target note	Yes	No	1	1	Aimed at note, pressed grip button first; paused and asked researcher 'How do I pick up the card?'. Prompted to try index trigger, then successfully picked up note.	Intervention: prompted to use index trigger; 1 error (grip button)
P02		3. Perform the intended note interaction	Yes	Yes	0	0	Held trigger steadily, translated note across field of view and inspected it.	Smooth manipulation once picked up
P02		4. Release / place the note correctly	Yes	Yes	0	0	Released trigger to place note down, checked that it stayed anchored in space.	Understood release interaction
P03	Moderate	1. Identify an interactive note	Yes	Yes	0	0	Spotted interactive note card immediately upon entering environment (<3 seconds).	High visual hierarchy
P03		2. Select / pick up the target note	Yes	Yes	0	0	Pointed laser pointer at note and pulled index trigger; immediate successful selection.	One-shot grab
P03		3. Perform the intended note interaction	Yes	Yes	0	0	Rotated and moved note closer and farther to test depth interaction smoothly.	Fluid control
P03		4. Release / place the note correctly	Yes	Yes	0	0	Released trigger to position note at chosen location; confirmed stable placement.	No placement issues
P04	Frequent	1. Identify an interactive note	Yes	Yes	0	0	Instantly identified the note card upon looking forward (<2 seconds).	Fast visual target acquisition
P04		2. Select / pick up the target note	Yes	Yes	0	0	Instantly targeted and grabbed note with index trigger (<2 seconds).	Highly proficient controller use
P04		3. Perform the intended note interaction	Yes	Yes	0	0	Freely moved, adjusted depth, and inspected card orientation seamlessly.	Zero hesitation
P04		4. Release / place the note correctly	Yes	Yes	0	0	Positioned note and released trigger decisively at target spot.	Task finished quickly

Overall observation — complete one row per participant

Select the best-matching overall behaviour; record mixed or changing behaviour in the task notes. Completion rate = successful tasks / 4.

Participant ID	Interaction understanding	Visual clarity	System feedback	Overall behaviour	Total successful	Total errors	Total assistance	Completion rate
P01	High	Clear	Clear	Smooth and autonomous	4	0	0	100%
P02	Moderate	Clear	Clear	Minor hesitation on Task 2 (button confusion), smooth thereafter	4	1	1	100%
P03	High	Clear	Clear	Confident and autonomous	4	0	0	100%
P04	High	Clear	Clear	Highly fluent and rapid	4	0	0	100%

Appendix B Interview Data

Figure. Design Evaluation 2 post-test interview results collected during the Week 9 Studio session. This table summarises the post-test interview responses from the same four participants who took part in the Week 9 Studio usability evaluation. The interview focused on interaction clarity, controller usability, system feedback, areas of confusion or desired improvement, and overall usability ratings. These responses were analysed together with the observation data to support the qualitative findings and design recommendations.

DECO7230 Design Evaluation 2 — Interview

Ask the four questions after testing. Record brief answers or exact quotes in the matching participant row.
Q1. Was it clear what you were supposed to interact with in the environment?
Q2. How easy or difficult was it to use the Quest controller to select and interact with the notes?
Q3. Was it clear when the system had recognised your interaction?
Q4. What was most confusing, and what one thing would you most like to improve?
Overall usability rating: 1 = Very difficult; 2 = Difficult; 3 = Neutral; 4 = Easy; 5 = Very easy.

Participant ID	Q1 Interaction target clarity	Q2 Controller ease	Q3 System recognition / feedback	Q4 Confusion / desired improvement	Overall usability rating (1–5)
P01	Very clear. The note card had distinct floating visual framing and depth that immediately distinguished it from the background environment.	Very easy. The laser ray pointing and index trigger pull felt natural and responsive.	Yes, very clear. There was an obvious visual outline/highlight when hovered and grabbed, confirming selection.	Placement alignment could be slightly improved; adding subtle haptic vibration upon grab and release would give better tactile confirmation.	5
P02	Clear. The target note was visually prominent and easy to locate in the scene.	Mostly easy, though slightly confusing initially. I tried pressing the side grip button first and had to ask how to pick it up. Once told to use the index trigger, it was straightforward.	Yes, once picked up, the note was firmly attached to the controller ray and followed hand movement reliably.	The controller button mapping was slightly ambiguous for a beginner. Adding an on-hover tooltip/hint like 'Pull Trigger to Pick Up' would prevent button confusion.	4
P03	Very clear. The visual hierarchy and color contrast made the interactive note stand out immediately.	Easy and intuitive. Aiming with the raycast and grabbing with the trigger was smooth with zero jitter.	Yes, the immediate highlight and attachment to the controller provided clear and unambiguous feedback.	Freehand depth positioning felt slightly imprecise. Having a soft snap-to-grid or surface anchor guideline would make placing notes neater.	5
P04	Extremely clear. Followed standard XR visual affordances and depth layering nicely.	Very easy. Standard Quest index trigger mapping is well-implemented and responsive.	Clear. The hover state and grab state transitions responded instantly with no perceptible lag.	A subtle audio cue (e.g. gentle click or chime upon grab/release) alongside visual highlight would polish the experience.	5